Beyond point-generated product–join constructions for projection-body volume

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Working research draft, 10 October 2026

Abstract

We give an explicit rational twelve-dimensional seed whose iterated products and joins have projection-volume root limit strictly between 2.861 and 2.8611. In combination with the prior bound $\Gamma_{\mathcal{C}} \le e^{1.049} < 2.855$ for the class generated from a point by products, joins, and affine equivalences, this proves a strict asymptotic separation between that entire class and unrestricted convex polytopes. Explicit finite product–join recipes and a fixed-block extension exceed every member of $\mathcal{C}_n$ for every $n \ge 9$. A simpler eight-dimensional seed already gives a limit above 2.8571. The geometric seed family, product–join calculus, recursive scheme, and comparison upper bound are attributed to prior work. The additional contribution is their use with specified rational seeds to obtain the stronger lower bounds and class separation. Independent exact geometry and recurrence certificates accompany the proof; global optimality and bibliographic priority are not asserted.

1 The separation problem and main result

For a full-dimensional convex body $K \subset \mathbb{R}^n$, use the normalization

$$h_{\Pi K}(u) = \mathrm{vol}_{n-1}(K \mid u^{\perp}), \qquad R_n(K) = \frac{\mathrm{vol}_n(\Pi K)}{\mathrm{vol}_n(K)^{n-1}}.$$

Here u is a unit vector and $K \mid u^{\perp}$ is its orthogonal projection. Write $g(n) = n^n/n!$ and $c_n = (n+1)g(n) = R_n(T_n)$, where T_n is an n-simplex. The ratio is invariant under invertible affine maps. For standard background on support functions, area measures, and affine convex geometry, see Schneider [16]. We omit the dimension subscript when it is unambiguous. The standard product and join are taken in independent coordinate spaces:

$$A * B = \mathrm{conv}\big((A,0,0) \cup (0,B,1)\big).$$

Let $\mathcal{C}$ consist of polytopes obtained from a zero-dimensional point by finitely many Cartesian products, joins, and invertible affine maps on their affine hulls. Rank-dropping affine maps are outside this definition. Set $\mathcal{C}_n = \{K \in \mathcal{C} : \dim K = n\}$ and

$$\Gamma_{\mathcal{C}} = \limsup_{n\to\infty} \sup_{K\in\mathcal{C}_n} R_n(K)^{1/n}.$$

The limit in fact exists by the spectral reduction in [11], but the limsup definition suffices below. Let U_n be the supremum of $R_n(K)^{1/n}$ over all full-dimensional convex bodies $K \subset \mathbb{R}^n$.

The unrestricted maximization problem has a longer history. Brannen [1] formulated the simplex maximum conjecture and disproved a different proposed upper bound for centrally symmetric

bodies. Lutwak, Yang, and Zhang [7] developed affine-invariant functionals for polytopes and derived projection-volume estimates. Saroglou [14] established sharp three-dimensional maxima within the classes of zonoids, cones, and double cones. These results concern the same volume ratio, but do not supply the whole-class comparison used here.

Feng, Hu, Liu, and Xu [4] already disprove the simplex upper bound in every dimension at least nine using polytopes with at most $n+2$ facets. OpenAI's product example [13] provides a particularly simple later counterexample. Subsequent work in the mxym repository gives the exact product–join calculus and a recursive construction with root limit between 2.8534 and 2.8535 [12, Section 9]. Its mixed Bellman bound [9] controls every finite expression in $\mathcal{C}$, not just that particular recursion. The question addressed here is whether the point-generated closure itself can attain the best unrestricted exponential rate.

Theorem 1. *There is an explicitly specified sequence of rational polytopes $P_n \subset \mathbb{R}^n$ such that*

$$\lim_{n\to\infty} R_n(P_n)^{1/n} = \beta, \qquad 2.861 < \beta < 2.8611.$$

Consequently, using the prior upper theorem [9],

$$\Gamma_{\mathcal{C}} < 2.855 < \beta \le \liminf_{n\to\infty} U_n.$$

There is also an explicit sequence E_n, given by finite product–join recipes and a fixed-block extension, satisfying

$$R_n(E_n) > \sup_{K\in\mathcal{C}_n} R_n(K) \qquad \text{for every } n \ge 9.$$

The sequence P_n can be chosen to satisfy the same strict inequality for every $n \ge 9$. No minimal-dimension claim is made.

Thus the advantage concerns an entire construction class. For example, the asymptotic ratio of $R_n(P_n)$ to its class supremum has lower root at least $2.861/2.855 > 1.0021$. This is an exponential separation; it is not a claim of optimality among all convex bodies. For context, Zhang's reverse polar projection inequality [17] and the Blaschke–Santaló inequality [8] give

$$R_n(K) \le \frac{\kappa_n^2 n^n}{\binom{2n}{n}}, \qquad \limsup_{n\to\infty} U_n \le \frac{\pi e}{2},$$

where κ_n is the volume of the Euclidean unit ball; see [15, Introduction, equation (3)] for the combined estimate. The limiting constant follows from Stirling's formula. A substantial unrestricted gap therefore remains.

2 Facet invariants and inherited operation rules

For a polytope K of dimension d and volume V, let v_i be its outward facet area-normal vectors and let ζ_i be the facet area times its support number. Equivalently, $\langle v_i, x\rangle = \zeta_i$ on facet i. Define the determinant sums

$$\mathcal{P}(K) = \sum_{|J|=d} |\det(v_j : j \in J)|, \qquad \mathcal{S}(K) = \sum_{|J|=d+1} |\det((v_j,\zeta_j) : j \in J)|.$$

Cauchy's projection formula gives $\mathcal{P}(K) = \mathrm{vol}_d(\Pi K)$: the projection body is a translate of the zonotope $\sum_i [0, v_i]$. The determinant expression for zonotope volume and its application to projection bodies are classical; see [7, Section 1] and [15, Section 3]. The lifted invariant of [12, Section 2] is

$$a(K) = \frac{\mathcal{S}(K)}{dV\mathcal{P}(K)}, \qquad H(K) = \frac{1}{a(K)}, \qquad Q(K) = \frac{a(K)R_d(K)}{g(d)}. \tag{1}$$

It is affine invariant and satisfies $1/(d+1) \le a(K) \le 1$. We use different symbols for the geometric tilt parameters below and the invariant a.

For positive-dimensional factors A, B of dimensions r, s, the exact identities in [12, Sections 3–5] are

$$R(A \times B) = R(A)R(B), \qquad a(A \times B) = \frac{ra(A) + sa(B)}{r+s}, \tag{2}$$

$$R(A * B) = \frac{g(r+s+1)}{g(r)g(s)} R(A)R(B)(a(A)+a(B)), \qquad a(A * B) = \frac{a(A)a(B)}{a(A)+a(B)}. \tag{3}$$

These rules apply to the nonsimplex seeds as well as to members of $\mathcal{C}$. They, and the balanced recursion used below, are imported prior results rather than new operation formulas.

3 An explicit rational seed

Let $n = 2m$ with $m \ge 1$, choose $0 < u_1 < \cdots < u_m < 1$, and put

$$b = (-u_m, \dots, -u_1, u_1, \dots, u_m), \qquad K(u) = \{x \ge 0 : \ \mathbf{1} \cdot x + b \cdot x \le 1, \ \mathbf{1} \cdot x - b \cdot x \le 1\}.$$

For rational u, this is a rational realization of the central two-facet section family underlying [4, Section 4]. We do not claim the family itself. The symmetry exchanging paired coordinates interchanges the two upper facets; it is not central symmetry of $K(u)$.

Proposition 2. *The polytope $K(u)$ has $n + 1 + m^2$ vertices and $n + 2$ facets. Put*

$$I(u) = \sum_{i=1}^{m} \frac{u_i^{2m-2}}{(1+u_i)\prod_{j \ne i}(u_i^2 - u_j^2)}, \qquad p(u) = n + 1 + \sum_{j=1}^{m} (2j-1)u_j, \qquad s(u) = 1 + \sum_{j=1}^{m} u_j. \tag{4}$$

Then

$$n! \operatorname{vol}_n(K(u)) = I(u), \qquad R_n(K(u)) = g(n)I(u)p(u), \qquad a(K(u)) = \frac{s(u)}{p(u)}. \tag{5}$$

In particular $Q(K(u)) = I(u)s(u)$.

Proof. The vertices are the origin, the n points $e_i/(1+|b_i|)$, and, for each pair $b_i = -v < 0 < b_j = w$,

$$\frac{w}{v+w} e_i + \frac{v}{v+w} e_j.$$

An active-set argument proves completeness: at a nonzero vertex one or both upper facets must be active, leaving at most one or two positive coordinates, respectively. In the latter case the signs must be opposite. Each coordinate hyperplane contains the origin and $n-1$ independent axis vertices. Each upper hyperplane contains a point with all coordinates positive and $b \cdot x$ of the required strict sign. Thus every listed inequality defines a facet.

On $b \cdot x \ge 0$, the upper constraint with $+b$ determines feasibility; the other upper constraint is redundant there. Change variables $y_i = (1+b_i)x_i$ and use the standard simplex with slack coordinate $y_0 = 1 - \sum_{i=1}^{n} y_i$. Its truncating coefficients are $t_0 = 0$ and $t_i = b_i/(1+b_i)$. The Jacobian is $\prod_i (1+b_i)^{-1}$ and $t_i - t_j = (b_i - b_j)/((1+b_i)(1+b_j))$ for $i, j \ge 1$. The simplex truncation identity in Appendix A, with these factors cancelled, gives

$$n! \operatorname{vol}_n(K(u) \cap \{b \cdot x \ge 0\}) = \sum_{b_i > 0} \frac{b_i^{n-1}}{(1+b_i)\prod_{j \ne i}(b_i - b_j)}.$$

The two halves have equal volume. Pairing $b_j = \pm u_j$ in the last denominator gives (4).

Write $V = \mathrm{vol}_n(K(u))$. Each upper facet vector is a positive multiple of $\mathbf{1} \pm b$. Reflection makes the two multiples equal. The divergence identity $\sum_i \zeta_i = nV$, with zero support on coordinate facets, makes each multiple $nV/2$. Balance of area-normal vectors then gives $-nVe_i$ on coordinate facet i. After division by nV, the horizontal generators and their lifted support coordinates are

$$(-e_i, 0), \qquad \left(\frac{\mathbf{1}+b}{2}, \frac{1}{2}\right), \quad \left(\frac{\mathbf{1}-b}{2}, \frac{1}{2}\right).$$

The horizontal n-minor sum is

$$1 + n + \frac{1}{2}\sum_{i<j} |b_i - b_j| = n + 1 + \sum_{j=1}^{m} (2j-1)u_j = p(u).$$

The lifted $(n+1)$-minor sum is $1 + \frac{1}{2}\sum_i |b_i| = s(u)$. Restoring the factors $(nV)^n$ and $(nV)^{n+1}$ in the two sums and using (1) proves (5). □

Our principal seed is the following body with 49 vertices and 14 facets:

$$K_0 = K\left(\frac{17}{100}, \frac{24}{100}, \frac{32}{100}, \frac{41}{100}, \frac{53}{100}, \frac{69}{100}\right). \tag{6}$$

In integer form its upper inequalities are

$$\sum_i (100 \pm B_i)x_i \le 100, \qquad B = (-69, -53, -41, -32, -24, -17, 17, 24, 32, 41, 53, 69).$$

Direct rational evaluation, independently checked from its geometry, gives

$$I_0 = \frac{1031929453906486710787861599001}{2386952337826268810834539955652}, \tag{7}$$

$$p_0 = \frac{768}{25}, \qquad s_0 = \frac{84}{25}, \qquad a_0 = \frac{7}{64}, \qquad R_0 = g(12)I_0\frac{768}{25}. \tag{8}$$

For comparison, the simpler seed $K(1/10, 2/10, 3/10, 4/10)$ in dimension eight has 25 vertices and

$$I_0 = \frac{231953}{378378}, \qquad p_0 = 14, \qquad s_0 = 2, \qquad a_0 = \frac{1}{7}.$$

Its initial ratio $R_8/c_8 = 231953/243243$ is below one. Nevertheless, the same recursive amplification already exceeds the asymptotic ceiling for $\mathcal{C}$. A seed's immediate projection-volume root therefore does not by itself describe its value under repeated operation rules.

4 Recursive amplification and exact tail control

Starting from either specified seed, apply the prior scheme of [12, Section 9]:

$$K_{j+1} = (K_j \times K_j) * (K_j \times K_j), \qquad d_j = \dim K_j, \quad R_j = R_{d_j}(K_j), \quad a_j = a(K_j). \tag{9}$$

The inherited identities give

$$d_j = \frac{(3d_0+1)4^j - 1}{3}, \qquad a_j = \frac{a_0}{2^j}, \qquad R_{j+1} = R_j^4 D_j, \quad D_j = 2a_j\frac{g(4d_j+1)}{g(2d_j)^2}. \tag{10}$$

Proposition 3 (Exact seed ranking at fixed dimension)**.** *For any two full-dimensional polytope seeds A, B of the same positive dimension d, the recursion (9) has positive root limits β_A, β_B and*

$$\left(\frac{\beta_A}{\beta_B}\right)^{3d+1} = \frac{a(A)R_d(A)^3}{a(B)R_d(B)^3}. \tag{11}$$

Thus maximizing the limiting rate in the fixed-dimensional family $K(u)$ is exactly equivalent to maximizing the rational objective

$$J(u) = I(u)^3 p(u)^2 s(u).$$

This ranking concerns the specified recursion, not every operation schedule.

Proof. The central-binomial bounds used below imply that D_j lies between two positive constants depending only on d, a_0. The logarithmic recurrence therefore gives a finite root limit for every seed. For equal-dimensional seeds their multiplier ratio is $a(A)/a(B)$ at every level, so exact iteration gives

$$\frac{R_j(A)}{R_j(B)} = \left(\frac{R_d(A)}{R_d(B)}\right)^{4^j} \left(\frac{a(A)}{a(B)}\right)^{(4^j-1)/3}.$$

Take d_jth roots and use $4^j/d_j \to 3/(3d+1)$ to prove (11). Finally $aR^3 = g(d)^3 I^3 p^2 s$ by (5). The factor $g(d)^3$ is constant at fixed dimension. □

Lemma 4. *For both seeds and every $j \geq 0$, one has $3 < D_j < 7$ and $R_j^{1/d_j} < 3$. The roots R_j^{1/d_j} strictly increase to a limit β with*

$$(3R_j^3)^{1/(3d_j+1)} < \beta < (7R_j^3)^{1/(3d_j+1)}. \tag{12}$$

Proof. We extend the elementary tail argument in [12, Lemma 9.1] to these seed states. For $q \geq 1$,

$$\frac{4^q}{2\sqrt{q}} \leq \binom{2q}{q} \leq \frac{4^q}{\sqrt{3q+1}}.$$

Both follow by induction from equality at $q = 1$. For the lower bound the squared induction comparison has difference $(2q+1)^2 - 4q(q+1) = 1$; for the upper bound it has difference $4(q+1)^2(3q+1) - (2q+1)^2(3q+4) = q$. Set $q = 2d_j$ and note that

$$\frac{g(2q+1)}{g(q)^2} = \frac{4^q}{\binom{2q}{q}} \left(1 + \frac{1}{2q}\right)^{2q}. \tag{13}$$

The last factor is strictly between 2 and e. Thus

$$4a_j\sqrt{6d_j+1} < D_j < 4ea_j\sqrt{2d_j}.$$

The closed dimension formula implies

$$a_j^2(6d_j+1) = a_0^2(6d_0+2-4^{-j}) \geq a_0^2(6d_0+1),$$

and $a_j^2(2d_j) < a_0^2(2d_0+2/3)$. The exact checks

$$16a_0^2(6d_0+1) > 9, \qquad 121a_0^2(2d_0+2/3) < 49, \qquad 7R_0^3 < 3^{3d_0+1} \tag{14}$$

hold for both seeds. The elementary bound $e < 11/4$ proves $D_j < 7$. Iterating $R_{j+1} < 7R_j^4$ gives

$$R_j^3 \le \frac{(7R_0^3)^{4^j}}{7} < \frac{3^{(3d_0+1)4^j}}{7} = \frac{3^{3d_j+1}}{7} < 3^{3d_j}.$$

Since $D_j > 3 > R_j^{1/d_j}$, the recurrence proves strict root increase. The upper bound by three proves convergence. Iteration of the logarithmic recurrence and the dimension formula yield the absolutely convergent series

$$\log \beta = \frac{\log R_j + \sum_{t\ge 0} 4^{-t-1} \log D_{j+t}}{d_j + 1/3}.$$

Using $3 < D_{j+t} < 7$ and $\sum_{t\ge 0} 4^{-t-1} = 1/3$ proves the two strict bounds in (12). □

Proposition 5. *For the twelve-dimensional seed,* $2.861 < \beta < 2.8611$. *For the eight-dimensional seed,* $2.8571 < \beta < 2.8573$.

Proof. The finite certificate stores the full exact rational state R_5 for each seed. The checker reconstructs it using (13), without importing the producer's factorial formula. It verifies by integer comparison

$$\ell^{3d_5+1} < 3R_5^3, \qquad 7R_5^3 < h^{3d_5+1},$$

with the following rational endpoints:

Seed dimension	d_5	ℓ	h
8	8533	28571/10000	28573/10000
12	12629	2861/1000	28611/10000

Together with Lemma 4, these finite inequalities certify the limits. No floating-point logarithm or numerically truncated infinite product is used in acceptance of the endpoints. □

For the asymptotic construction in Theorem 1, take the twelve-dimensional family. For $n \ge 144$, choose the largest j with $d_j \le \sqrt{n}$, put $k = \lfloor n/d_j \rfloor$, $r = n - kd_j$, and set

$$A_n = K_j^k \times T_r.$$

Here K_j^k denotes a Cartesian power; the factor T_0 is omitted. For $n < 144$ use $A_n = T_n$. Then $j \to \infty$, $0 \le r < d_j \le \sqrt{n}$, and $kd_j/n \to 1$. Product multiplicativity gives

$$\frac{\log R_n(A_n)}{n} = \frac{kd_j}{n}\frac{\log R_j}{d_j} + \frac{\log c_r}{n}.$$

Set $c_0 = 1$. The bound $c_r \le (r+1)e^r$ makes the last term tend to zero, so the required all-dimensional limit is β. This padding argument is the one used for the prior recursive family in [12, Corollary 9.3].

5 The imported upper bound and separation from dimension nine

The mixed Bellman theorem [9] states, for every $K \in \mathcal{C}$, with $D = n + 1$ and $H = 1/a(K)$,

$$\log Q(K) \le \alpha\left(D - \frac{H^2}{D}\right) + \delta\left(D - \frac{H^4}{D^3}\right), \qquad \alpha = \frac{87}{2000}, \quad \delta = \frac{11}{2000}. \tag{15}$$

This is an imported theorem, not a new inequality of this note. Its pinned exact certificate was replayed successfully: 19,900 finite state rectangles, 15,568 tail cells, and the analytic large-dimension constants. The complete English argument was also inspected. Exact replay checks its finite proof obligations; it does not replace review of its analytic argument.

Discarding the dependence on H gives the coarse bound $R_n(K) < 1.051(n+1)(2.8548)^n$. Combined with a single block and simplex padding, this gave the earlier sufficient threshold 15500. Retaining that dependence yields a much smaller comparison bound. Write

$$f_n(H) = \log H + \alpha(D - H^2/D) + \delta(D - H^4/D^3).$$

Then $R_n(K) \le g(n) \exp f_n(H)$. For every rational $h > 0$, $f_n''(H) \le -2\alpha/D$ implies

$$\sup_{K \in \mathcal{C}_n} R_n(K) \le g(n) h \exp\left[\alpha(D - h^2/D) + \delta(D - h^4/D^3) + \frac{D}{4\alpha}\left(\frac{1}{h} - \frac{2\alpha h}{D} - \frac{4\delta h^3}{D^3}\right)^2\right]. \tag{16}$$

Indeed, the quadratic upper Taylor bound at h is maximized by completing the square. Rational centres near its stationary point give tight bounds without solving an algebraic equation in proof acceptance.

Lemma 6. *For every $n \ge 49$,*

$$\sup_{K \in \mathcal{C}_n} R_n(K) < \frac{9}{10}\left(\frac{7137}{2500}\right)^n. \tag{17}$$

Proof. Drop the nonpositive quartic term in $f_n(H)$ and use $xe^{-x} \le e^{-1}$, with $x = 2\alpha H^2/D$. This gives

$$R_n(K)^2 \le \frac{1000}{87}(n+1)g(n)^2 \exp\left(\frac{49}{500}(n+1) - 1\right).$$

The sequence $B_n = (n+1)g(n)^2 e^{-2n}$ strictly decreases. Its consecutive ratio is $(n+2)(1+1/n)^{2n}/((n+1)e^2)$, which is below one since

$$2n\log(1+1/n) < 2 - \frac{1}{n+1}, \qquad \log(1+1/(n+1)) < \frac{1}{n+1}.$$

The first inequality follows by integrating $t/(1+t) \ge t/(1+x)$ on $0 \le t \le x$, with $x = 1/n$. Consequently, for $n \ge 49$,

$$R_n(K)^2 \le \frac{1000}{87} B_{49} e^{-451/500} e^{(1049/500)n}.$$

Rational Taylor bounds certify

$$\frac{1000}{87}\, 50 g(49)^2 e^{-98-451/500} < \frac{81}{100}, \qquad e^{1049/1000} < \frac{7137}{2500}.$$

Taking positive square roots proves the claim. In particular $\Gamma_{\mathcal{C}} < 2.855$, as used in Theorem 1. □

Two additional rational members of the inherited seed family suffice for the finite witnesses. The ten-dimensional seed is

$$S_{10} = K(16/1000, 51/1000, 91/1000, 142/1000, 217/1000).$$

The nine-dimensional seed S_9 is a pyramid over $K(1/100, 2/100, 3/100, 5/100)$ in dimension eight; equivalently, append one zero coefficient to its paired tilt vector. It has 26 vertices and 11 facets, whereas S_{10} has 36 vertices and 12 facets. The calculus in (H, Q) extends to a point with the formal values $H = Q = 1$:

$$(H, Q)(A * B) = (H_A + H_B, Q_A Q_B), \tag{18}$$

$$H(A \times B) = \frac{r+s}{r/H_A + s/H_B}, \qquad Q(A \times B) = Q_A Q_B \frac{g(r)g(s)}{g(r+s)} \frac{sH_A + rH_B}{r+s}. \tag{19}$$

The point case in (18) is the ordinary pyramid formula; products with a point can be omitted. These are rewritings of the prior operation rules, not new geometric identities.

Proposition 7. *The accompanying 420-node rational operation certificate specifies polytopes W_n for $9 \le n \le 245$, built from points, S_9, S_{10}, and K_0 by products and joins. Let $L = K_2$, of dimension 197. Define*

$$E_n = \begin{cases} W_n, & 9 \le n \le 48, \\ L^k \times W_s, & n \ge 49, \quad s = 49 + ((n-49) \bmod 197), \quad k = (n-s)/197. \end{cases}$$

Then $R_n(E_n) > \sup_{K \in \mathcal{C}_n} R_n(K)$ for every $n \ge 9$. The factor L^0 is omitted.

Proof. For $n = 9$, the prior exact Pareto classification gives $\sup_{K \in \mathcal{C}_9} R_9(K) = c_9$ [10, Theorem 1]. Our checker replays this small instance: starting with $(H, Q) = (1, 1)$, it enumerates every product and join of smaller-dimensional Pareto states, then discards coordinatewise dominated states. Both outputs in (18)–(19) increase in both input coordinates at fixed dimensions, so induction proves completeness. The replay checks 261 candidates, retaining 50 states across dimensions zero through nine. The independent geometry check gives

$$\frac{R_9(S_9)}{c_9} = \frac{20811283583}{20797677600} > 1.$$

Set $W_9 = S_9$. The geometric family and the dimension-nine simplex counterexample phenomenon are already prior work [4].

For $10 \le n \le 48$, the certificate specifies W_n and a rational centre h_n. Reconstructing every operation state and applying rational upper Taylor intervals for the exponential in (16) gives the strict comparison in each of these 39 dimensions. For each of the 197 integers $49 \le s \le 245$, it instead certifies

$$R_s(W_s) > \frac{1001}{1000} \frac{9}{10} \left(\frac{7137}{2500} \right)^s.$$

The dimension ranges and certified multiplicative margins are:

Dimensions	Class upper bound	Witness / upper bound
9	c_9, exact Pareto replay	$> 4001/4000$
10–48	Rational tangent bounds	$> 4001/4000$
49–245	$(9/10)(7137/2500)^n$	$> 1001/1000$

These are verified lower margins, not approximations to sharp extrema. For example, the recipes choose $W_{10} = S_{10}$, W_{18} as six successive pyramids over K_0, and $W_{49} = K_1$. The finite recipes are explicit construction witnesses; completeness of the search that discovered them is unnecessary.

Finally the exact recurrence gives

$$R_{197}(L) > \frac{5}{4}\left(\frac{7137}{2500}\right)^{197}.$$

Product multiplicativity and Lemma 6 now prove the claim for every $n = s + 197k \geq 49$. This covers every residue and every larger dimension, with no finite truncation of the conclusion. The block retains its explicit 224-facet description and has 46,118,408 vertices, counted by the recursion rather than enumerated. □

To obtain one sequence satisfying both assertions of Theorem 1, set $E_n = T_n$ for $n < 9$ and choose P_n to be whichever of A_n, E_n has larger R_n, breaking ties in favour of A_n. This is an exact rational comparison. Since $R_n(E_n)^{1/n} \to R_{197}(L)^{1/197} < \beta$ by strict root increase, whereas $R_n(A_n)^{1/n} \to \beta$, the selected sequence has limit β and strictly exceeds the entire class in every dimension $n \geq 9$.

6 Exact verification

The accompanying standard-library Python package contains the four exact seeds, their vertex lists and oriented triangulations, the full rational recurrence states, and separate geometric and growth checkers. The producers use pulling triangulations; the standard geometric framework is described in [3]. The checkers validate the resulting oriented chains independently of their construction. For the twelve-dimensional seed, the geometric checker solves all 91 active systems, verifies 924 positively oriented simplices, cancels all interior boundary faces, and verifies multiplicity one at a generic rational interior point. The analogous counts for the eight-dimensional seed are 45 active systems and 70 simplices. All uncancelled faces lie on the body's boundary. Crossing an interior codimension-one face therefore leaves the positive-chain multiplicity unchanged, and a generic interior path avoids faces of codimension two. Multiplicity is constant in the interior; its checked value one proves coverage almost everywhere with no interior overlap.

The checker obtains volumes from simplex determinants and facet vectors from boundary cofactors, then sums every horizontal and lifted maximal minor. Thus it checks both R_0 and a_0 without assuming Proposition 2. The recurrence checker imports neither the geometric producer nor its formula module and uses central binomial coefficients instead of factorial ratios. It checks (14), the rational limit endpoints and the original dimension-197 block comparison. A separate finite-separation checker reconstructs all 420 witness nodes, replays the exact dimension-nine class frontier, checks the 39 rational tangent bounds and all 197 residue witnesses, and verifies the finite constants in the analytic all-dimensional extension. The new seeds have 55 and 66 active systems and 70 and 252 oriented simplices, respectively. Explicit exceptions keep the mathematical checks active under optimized Python. The 39 finite comparisons are also cross-checked using rational atanh intervals for logarithms, without reusing the exponential-series code. The dimension-197 block also has an explicit inequality certificate. A separate checker compares its 224 inequalities with a flattened description using sixteen seed copies, four inner join coordinates, and one outer join coordinate. It verifies a strict interior point. This checks equivalence to the recursive expression; it does not enumerate the block's vertices or independently recompute its volume.

Run `python verify_all.py` for all new checks and sixteen unit tests. The runner writes regenerated certificates and reports to temporary files and leaves the delivered files unchanged. Empty or incomplete delivery manifests and mathematically false comparison constants are rejected. The command `python replay_prior.py` fetches byte-pinned external upper-bound inputs and replays their checker. This optional step needs network access on first use. The delivery includes its

successful replay report and the input manifest. Formula and geometric data, unlike PDF build metadata, regenerate byte-identically in the tested environments.

7 Discussion

The separation in Theorem 1 identifies a limitation of the point-generated class rather than a failure of products and joins themselves. The same inherited operations amplify the new seeds beyond the upper rate available to every finite expression in $\mathcal{C}$. Thus changing the initial geometric input can matter even when the subsequent construction rules remain fixed. The result compares complete construction classes; exceeding one previously known recursive sequence alone would not establish this conclusion.

The eight-dimensional example illustrates why immediate seed performance can be misleading. Its ratio R_8/c_8 is below one, yet its recursively amplified rate exceeds the ceiling for $\mathcal{C}$. The lifted invariant a records information used by joins that the initial value of R alone does not capture. Proposition 3 makes this observation quantitative: at a fixed dimension and for the specified recursion, the correct seed objective is aR^3, or equivalently I^3p^2s in the stated tilt family. This supplies an exact target for further seed search. It does not justify comparing different dimensions by that objective or claiming that the balanced recursion is the best possible schedule.

The asymptotic advantage is small at the level of a single root, but it produces an exponential ratio of volumes as dimension grows. The finite construction now separates in every dimension at least nine. The reduction from the earlier sufficient threshold 15500 comes from retaining the invariant dependence in the class envelope and replacing a uniform simplex padding loss by finitely many checked product–join witnesses. It does not improve the imported asymptotic class ceiling or the certified value of β. Different finite and asymptotic objectives favour different seeds. The threshold nine is not proved minimal: the present result makes no new assertion about unrestricted maximizers in dimensions four through eight. Nor does the lower rate approach a known unrestricted optimum: the classical upper root $\pi e/2$ leaves a substantial gap.

Explicitness also needs a computational qualification. The block L has a compact description by 224 facet inequalities but more than 46 million vertices. Its certified ratio is obtained from the checked seed geometry and the inherited operation identities, rather than by enumerating its vertices or constructing its projection body directly. The package therefore makes these particular witnesses and their comparison independently checkable; it does not provide an efficient algorithm for optimizing projection-body volume over arbitrary polytopes.

The central geometric family is prior work [4], as are the lifted invariant, operation rules, recursion, and tail method [12]. The whole-class upper theorem [9] is an essential external dependency. Its exact finite obligations were replayed, but this does not formalize its analytic argument or constitute external peer review. The contribution relative to the inspected sources is the specified rational seeds, their certified amplified rates, and the resulting class separation and effective comparison. The supplementary literature audit does not establish bibliographic priority.

Further progress could take several forms. A rigorous global search for the objective I^3p^2s would distinguish a good rational seed from an optimum within a precisely stated family. Enlarging that family, changing seed dimension, or allowing unbalanced schedules may improve the lower rate. Sharper class envelopes may reduce the finite certificate or improve separation margins. Moving below dimension nine would require a new unrestricted counterexample, not merely a smaller padding constant. These optimization problems are not resolved by the present certificates. External mathematical review and proof-assistant formalization would strengthen confidence in both the new argument and its imported dependencies.

8 Conclusion

An explicit rational twelve-dimensional seed, amplified by the prior product–join recursion, gives a projection-volume root limit strictly between 2.861 and 2.8611. Combined with the attributed upper theorem for the point-generated class, this proves a strict exponential separation from that entire class. A finite rational operation certificate and a repeating 197-dimensional block give a comparison in every dimension $n \geq 9$. One explicit sequence satisfies this finite separation while retaining the stronger root limit.

The examples show that the initial seed and its lifted invariant are material to recursive growth. Exact geometry, recurrence certificates, and a finite comparison make the claims reproducible without treating numerical search as proof. The unrestricted extremal rate, optimal seed and schedule, minimal separation dimension, and global bibliographic priority remain open.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

Data and code availability

The manuscript source, PDF, exact seed coordinates and triangulations, dimension-197 facet inequalities, finite operation recipes, recurrence certificates, verification programs, source audit, and reproduction instructions are publicly available in the accompanying GitHub repository [5]. The versioned release `v1.1.0` identifies the edition accompanying this manuscript. Original Python code and generated certificate data are provided under the MIT license; manuscript rights are stated separately.

From the release's `reproduction/` directory, run `python -B verify_all.py -regenerate` with Python 3.11 or newer. All checks of the new constructions use only the standard library and run without network access. The optional flag `-with-prior` fetches and replays the byte-pinned external upper-theorem package; its first use requires network access. The delivery manifest records SHA-256 hashes, and verification leaves delivered files unchanged. The release includes the successful external replay record. External inputs remain attributed and are fetched from their pinned upstream locations rather than redistributed as part of this package.

A A simplex truncation identity

The following is a standard simplex section formula; see Lasserre [6, Theorem 2.2]. We include a proof in the normalization used here, using the Genocchi–Hermite representation of divided differences [2, Section 9]. Let $t_0, \dots, t_n$ be distinct real numbers, and let $\Delta_n = \{y_i \geq 0 : \sum_{i=1}^n y_i \leq 1\}$, with $y_0 = 1 - \sum_{i=1}^n y_i$. Then

$$\frac{\mathrm{vol}_n\{y \in \Delta_n : \sum_{i=0}^n t_i y_i \geq 0\}}{\mathrm{vol}_n(\Delta_n)} = \sum_{t_i > 0} \frac{t_i^n}{\prod_{j \neq i}(t_i - t_j)}. \tag{20}$$

Here a zero node contributes zero. For completeness, the elementary divided-difference integration identity is

$$\int_{\Delta_n} f^{(n)}\left(\sum_{i=0}^n t_i y_i\right) dy = \sum_{i=0}^n \frac{f(t_i)}{\prod_{j \neq i}(t_i - t_j)}.$$

One proves it by induction using the fundamental theorem of calculus in y_n. Multiplication by $t_n - t_0$ turns the integral into the difference of the two $(n-1)$-simplex integrals on the faces $y_0 = 0$ and $y_n = 0$. The induction hypothesis gives exactly the divided-difference recurrence; the initial case $n = 0$ is evaluation at a point. Apply this identity to $f(t) = t_+^n/n!$. Its nth derivative is the positive-half-line indicator almost everywhere; the integration identity still holds by the same fundamental-theorem argument, or by smooth approximation. Since $\mathrm{vol}_n(\Delta_n) = 1/n!$, equation (20) follows.

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